HOMEWORK DAY 3 – *Logarithmic function §6.3*

1. §6.3: 3. *(a) Answer:* $\log_3 81 = 4$ *since* $3^4 = 9^2 = 81$

2. §6.3: 4.

3. §6.3: 5.

4. §6.3: 6.

5. §6.3: 10(b).

6. §6.3: 43.

7. §6.3: 44.

8. §6.3: 45.

9. Sketch the graphs of the following functions.

(a) $f(x) = \ln(x)$

(b) $f(x) = \ln(1/x)$

(c) $f(x) = \ln(x - 1)$

(d) $f(x) = \ln|x|$

(e) $f(x) = \ln(x + c)$, $0 < c < 1$

10. Solve the following inequalities and equations for x.

(a) $\ln x < 0$

(b) $\ln x > 1$

(c) $\frac{e^x}{1-x} = 0$

(d) $\ln(e^x - 3) = 2$

(e) $\ln x + \ln(x + 1) = 0$

11. Evaluate the limits.

(a) $\lim_{x \to 1^+} \ln(x-1)$

(b) $\lim_{x \to 1^-} \ln(x-1)$

(c) $\lim_{x \to 0} \ln(\cos x)$

(d) $\lim_{x \to 0^+} \ln(\sin x)$

HOMEWORK DAY 4 – *Derivatives of* $\ln x, \log_b(x), b^x$ *§6.4*

12. §6.4: 2.

13. §6.4: 4.

14. §6.4: 5.

15. §6.4: 6.

16. §6.4: 7.

17. §6.4: 11.

18. §6.4: 13.

19. §6.4: 17.

20. §6.4: 47 (logarithmic differentiation)

21. §6.4: 52 (logarithmic differentiation)

22. Consider the function $f(x) = x \ln x$

    (a) State the domain of f.

    (b) Find all intercepts.

    (c) Find where f is increasing/decreasing.

    (d) Find where f is concave up/down.

    (e) Find $\lim_{x\to\infty} f(x)$.

    (f) Using a table of values, estimate $\lim_{x\to 0^+} f(x)$. (Include your table at the margin of this page.)

    (g) Use the above information to sketch a graph of $f(x)$

23. §6.4: 76.

24. §6.4: 81.

25. §6.4: 84.

26. §6.4: 86.

HOMEWORK DAY 5 – *Inverse trig functions §6.6*

27. *Values of inverse trig functions.* Find the following values (without using a calculator).

    (a) $\sin^{-1}\left(\frac{1}{2}\right)$.

    Answer : $\sin^{-1}\left(\frac{1}{2}\right) = \alpha$ where $\sin(\alpha) = \frac{1}{2}$, $\alpha \in [-\pi/2, \pi/2]$ (range of $\sin^{-1}$). So $\alpha = \pi/6$.

    (b) $\cos^{-1}\left(\frac{1}{2}\right)$

    (c) $\sin^{-1}(1)$

    (d) $\cos^{-1}(-1)$

    (e) $\tan^{-1}(-\sqrt{3})$

    (f) $\sec^{-1}(2)$

    (g) $\cos^{-1}(\sqrt{3}/2)$

28. *Graphs of inverse trig functions.* Sketch graphs of the following functions on two side-by-side plots (make a separate plot for each function).

    (a) $y = \sin x, -\pi/2 \leq x \leq \pi/2$ and $y = \sin^{-1} x$.

    (b) $y = \cos x, 0 \leq x \leq \pi$ and $y = \cos^{-1} x$

    (c) $y = \tan x, -\pi/2 < x < \pi/2$ and $y = \tan^{-1} x$

29. By drawing the appropriate triangle, find

(a) $\cos(\sin^{-1} x)$ (Hint: draw a right triangle with angle $\alpha = \sin^{-1} x$, that is $\sin\alpha = \frac{x}{1} = \frac{\text{opp}}{\text{hyp}}$)

(b) $\tan(\sin^{-1} x)$